

Course No: **CSE-209**
Full Marks: **10**

1. Answer **Any Two (02) questions** out of **Three (03)**

- | Item | Value | Weight |
|------|-------|--------|
| 1. | 5 | 2 |
| 2. | 20 | 1 |
| 3. | 15 | 7 |
| 4. | 10 | 3 |
| 5. | 40 | 8 |
| 6. | 25 | 4 |

-

Page 1 of 3

- c) A file has the following characters along with their frequencies: 2.5

d : 7, e : 11, c : 5, a : **A**, m : **B**, p : **C**, q : **D**

A = alphabetic order of the 1st letter of your first name

B = alphabetic order of the 2nd letter of your first name

C = alphabetic order of the 3rd letter of your first name

D = absolute difference between alphabetic order of the 1st letter and 3rd letter of your first name (If the value of D is zero (0), then consider D = 8)

Draw Huffman coding tree and compute the optimal Huffman code of each character mentioned above.

N.B.: Make sure you write your full name properly. Also write down the revised frequencies of the characters before your solution.

Answer Any One (01) out of Two (02)

2. a) i) Explain which approach (Divide & Conquer, Greedy, or Dynamic Programming) is applicable to solve the following problem and why: print the first index of an input number in a given sorted array in less than $O(n)$ time; print "Not Found" if it is not found. 1

Sample input/output:

Input: Array = [1, 5, 5, 5, 7, 7, 7, 7, 8, 9, 9, 9, 9, 9, 9, 9, 11], Search Key = 9

Output: First index of 9 is: 10

- (ii) The MergeSort algorithm we have learned is called 2-way MergeSort because it divides the given array into two almost equal sub-arrays. In 5-way MergeSort, the given array is divided into 5 equal (almost) sub-arrays and they are recursively sorted and then combined/merged to get the sorted array. Represent the time complexity of 5-way MergeSort algorithm using a recurrence relation and write it using Big-Theta notation. (Hint: The combine step merges a total of n elements, taking $O(n)$ time.) 2
- iii) What are the properties of a greedy algorithm? Write and explain that the **activity selection problem** satisfies all those properties. 2

- b) Let's assume you are the owner of **8** factories. Now, to accelerate the business you have to establish a communication network among your factories. You can communicate from one factory to another if and only if **they have direct connection** or **they are connected via one or more factories**. The cost matrix between each pair of factories is given below.

$$\text{cost}[][] = \begin{bmatrix} \infty & 300 & 600 & \infty & \infty & \infty & \infty & \infty \\ 300 & \infty & \infty & 400 & \infty & \infty & \infty & 600 \\ 600 & \infty & \infty & \infty & 400 & 200 & \infty & \infty \\ \infty & 400 & \infty & \infty & 600 & \infty & 300 & \infty \\ \infty & \infty & 400 & 600 & \infty & 600 & \infty & \infty \\ \infty & \infty & 200 & \infty & 600 & \infty & \infty & \infty \\ \infty & \infty & \infty & 300 & \infty & \infty & \infty & 400 \\ \infty & 600 & \infty & \infty & \infty & \infty & 400 & \infty \end{bmatrix}$$

Here, **cost[i][j]** represents the establishment cost of a direct connection between **factory_i** to **factory_j**.

Example,

$\text{cost}[1][2]=300$, which means it is possible to establish a direct connection between factory_1 and factory_2 and this establishment cost will be 300 Lac.

$\text{cost}[1][4]=\infty$, which means it is impossible to establish a direct connection between factory_1 and factory_4 .

Now answer each of the followings.

- i. Which algorithmic strategy will you **select** to solve this problem. 1
- ii. **Calculate** the minimum amount of cost that will be required to establish the communication network, so that it is possible to communicate between any two factories. 2
- iii. **Distinguish** between dynamic programming and greedy approach. 2



Jahangirnagar University
Department of Computer Science and Engineering
 2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2020

Course Title: Electronic Devices and Circuits-II
 Time: 45 minutes

Course No: **CSE-207**
 Full Marks: **10**

Answer any Two (02) out of Three (03).

[Each question carries equal marks. Figures in the right margin indicate marks.]

1. a) Calculate the dc voltages and currents in the circuit of Figure-1. 2.5

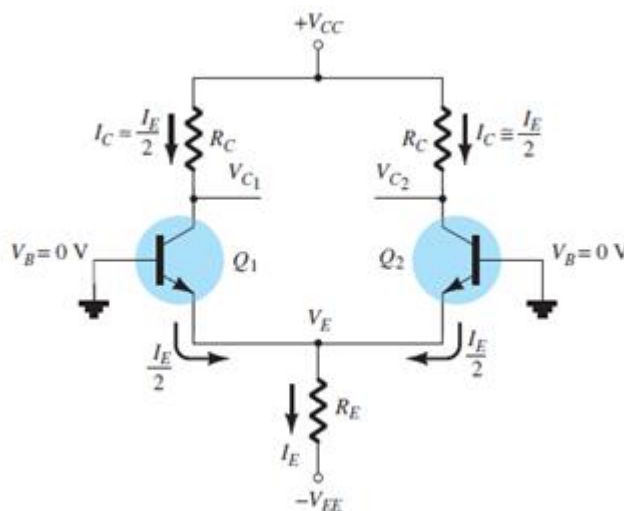


Figure-1

b) Determine quiescent point I_{CQ} and V_{CEQ} of the following circuit. Show them on the intersection of characteristics curve and load line.
 In the given figure, $R = (250 + X) \text{ k}\Omega$.
 Here, $X = (\text{Your Exam Roll mod } 10) + 1$. If your Exam Roll = 123456, Then $X = (123456 \text{ mod } 10) + 1 = 6 + 1 = 7$.

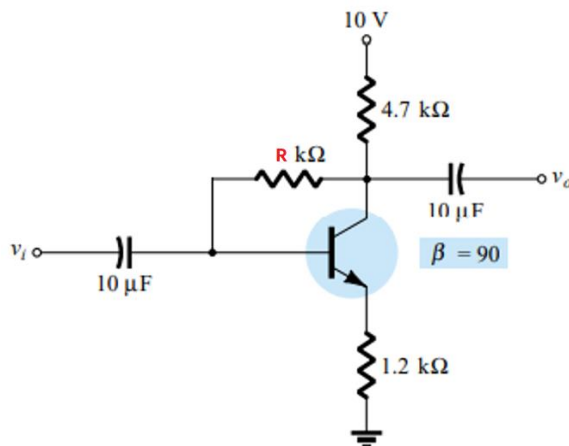
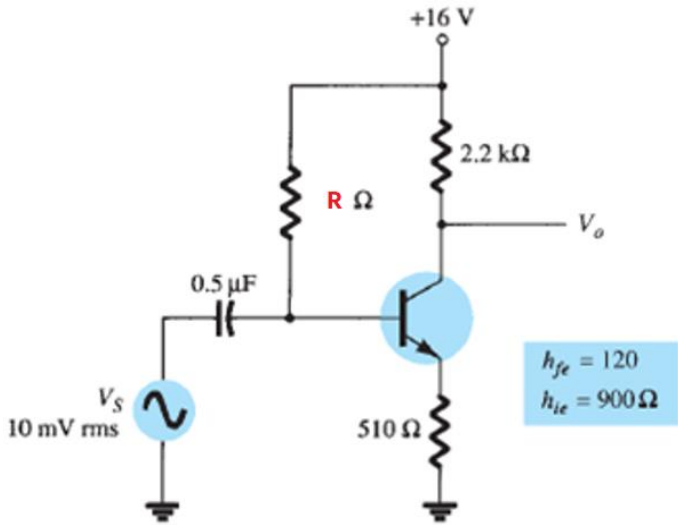


Figure-2

	c)	(i) Give the signification of CMRR of an OPAMP. If differential gain, $A_d = 120 + X$ and common-mode gain $A_c = 9$ then determine CMRR in dB. Here, $X = (\text{Your Exam Roll mod } 10) + 1$. If your Exam Roll = 123456, Then $X = (123456 \text{ mod } 10) + 1 = 6+1=7$. (ii) Give Barkhausen criteria of sustaining oscillation with circuit diagram.	2.5
--	----	--	-----

Answer Any One (01) out of Two (02)

[Each question carries equal marks. Figures in the right margin indicate marks.]

2.	a)	i) Explain how an op-amp acts as an integrator. ii) Illustrate how an op-amp is used as a multivibrator.	2.5
			2.5
	b)	Determine the voltage gain of the circuit of the Figure-3. Here, $R = (470 + X) \Omega$ Where, $X = (\text{Your Exam Roll mod } 10) + 1$. If your Exam Roll = 123456, Then $X = (123456 \text{ mod } 10) + 1 = 6+1=7$.  Figure-3	5



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2020

Course Title: Numerical Methods
Time: 1 hour

Course No: **CSE-205**
Full Marks: **10**

Answer any Two (02) out of Three (03).

[Each question carries equal marks. Figures in the right margin indicate marks.]

1. a) Solve the following sets of equations using Gauss-Seidel method: 2.5

$$\begin{aligned} 2x_1 + x_2 + x_3 &= 5 \\ 3x_1 + 5x_2 + 2x_3 &= 15 \\ 2x_1 + x_2 + 4x_3 &= 8 \end{aligned}$$

- b) Use two Segment Trapezoidal Rule to find the area under the curve: 2.5

$$f(x) = \frac{300x}{1 + e^x} \quad \text{from } x = 0 \text{ to } x = 10$$

- c) Find out the root of the equation $4x^3 - 2x - 6 = 0$ using bisection method. 2.5

Answer Any One (01) out of Two (02)

[Each question carries equal marks. Figures in the right margin indicate marks.]

2. a) i) Derive the general form of Lagrange interpolation polynomial. 2.5

- ii) Derive the Lagrange interpolation polynomial for the following data set. 2.5

i	0	1	2	3
x_i	0	1	2	3
$e^x - 1$	0	1.7183	6.3891	19.0855

- b) The velocity of a rocket is given by

$$v(t) = 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 2100t} \right] - 9.8t, \quad 0 \leq t \leq 30$$

Where v is given in m/s and t is given in seconds.

- i) Use forward difference approximation of the first derivative of $v(t)$ to calculate the acceleration at $t=16s$. Use a step size of $\Delta t=2s$. 2.5

- ii) Find the exact value of the acceleration of the rocket. 2.5

-----0-----
Good Luck

Department of Computer Science and Engineering
 Jahangirnagar University
2nd Year 1st Semester BSc (Hons.) Examination -2020
 Course No # **CSE-201**, Course Title: **Mathematics III**
 Full Marks: **10**, Time: **45** minutes
 [Answer any **four** questions]

1.	Find the angle between the surfaces $x^2 + y^2 + z^2 = 12$ and $z = x^2 + y^2 - 7$ at the point $(1, 5, 0)$.	[2.5]
2.	Find a constants a, b, c so that $\underline{v} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational.	[2.5]
3.	Evaluate $\oint_C \frac{e^{2z}}{(z-1)^2(z-3)} dz$; if C is the circle $ z = 2$.	[2.5]
4.	Using Cauchy's residue theorem evaluate $\int_0^\infty \frac{\sin 2x}{x} dx$.	[2.5]
5.	Find the Fourier integral of $f(x) = \begin{cases} x ; & -\pi \leq x \leq \pi \\ 0; & x > \pi \end{cases}$	[2.5]
6.	Let $f(t) = t$, $0 < t < 3$ is a periodic function of period 3. Then sketch $f(t)$ and find $L\{f(t)\}$.	[2.5]



Jahangirnagar University
Department of Computer Science and Engineering
_2nd _ Year 1st __ Semester B.Sc. (Hons.) Final Examination -2020

Course Title: Computer Ethics and Cyber Law
Time: 1 hr

Course No: CSE-203
Marks: 10

Answer Any two questions out of Three.

1. a) i. Should political speech be more rigorously protected than other forms of expression? **Illustrate** your point of view from the subjective point of view of computer ethics and cyber law. 1
- ii. **Identify** the key elements of a multilayer process for managing security vulnerabilities based on the concept of reasonable assurance. 1.5
- b) i. **Choose** the steps in order to secure a server. 1
- ii. “Malicious hacker takes over computers on the Internet and causes them to flood a target site with demands for data and other small tasks”. 1.5
Predict the types of attack it can be.
- c) **Classify** the areas covered by term intellectual property. Hence, **predict** the reasons behind companies so concerned about protecting intellectual property. 2.5

Answer Any One question out of Two.

2. a) i. You received an email from your Agrani bank telling you that there is a problem with your bank account. The email provided instructions and a link so that you can log into your account and fix the problem. 2.5
Point out your action in the given scenario.
- ii. There is the case that happened in my computer lab. A colleague of mine used her gmail account at a computer lab on campus. She ensured that her account was not left open before she left the lab. Someone came after her and used the same browser to re-access her account and they started sending threatening emails from it. 2.5
Examine the scenario and **predict** possible consequences with respect to computer ethics and cyber law.
- b) “You have joined into a software company. You have access to everything including the database and codebase. Your company provided you a high configuration laptop for official work. Your working hour is 9:00 Am to 5:00 PM. But you have to work more sometimes.”
- Analyze** the code of ethics for information technology professionals in the above scenario. Hence **Examine** the term cyber space. 2.5
- Analyze** the term computer ethics and **point out** your desired activities with respect to the top 10 commandments of computer ethics. 2.5